

Digital Image Processing

Ch 2. Digital Image Fundamentals

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Image Fundamentals

- Human eye and adaptation
- Sampling and Quantization
- Basic Relationships between pixels
- Imaging Geometry
- Photographic film

Structure of the Human Eye

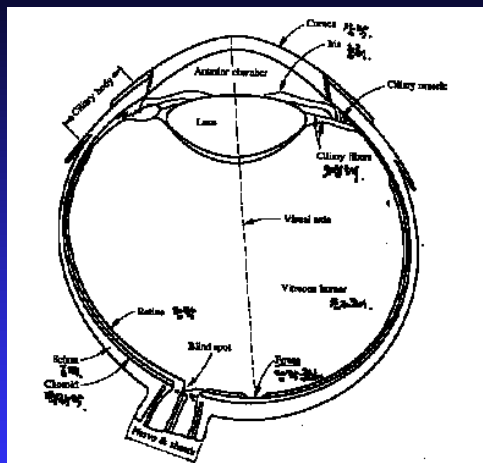


Figure 2.1 Simplified diagram of a cross section of the human eye

Structure of the Human Eye

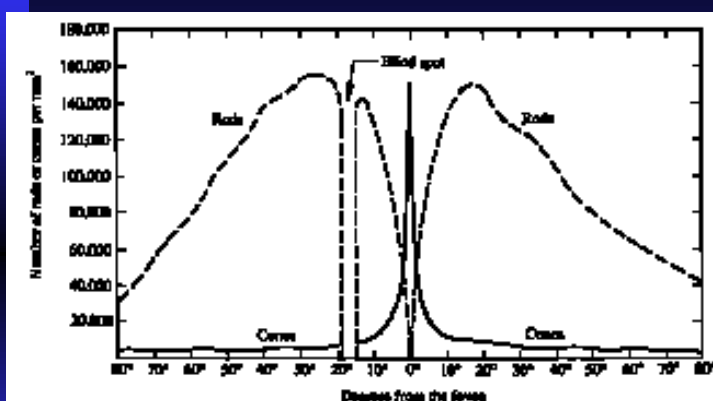


Figure 2.2 Distribution of rods and cones in the retina (Adapted from Graham [1965])

Brightness Adaptation

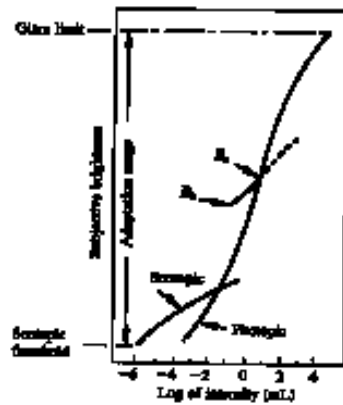


Figure 2.4 Range of subjective brightness sensations showing a particular adaptation level.

Brightness Discrimination

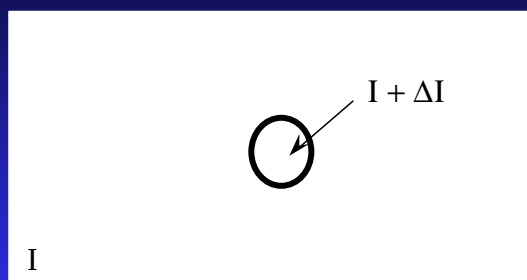


Figure 2.5 Basic experimental setup used to characterize brightness discrimination

Brightness Discrimination

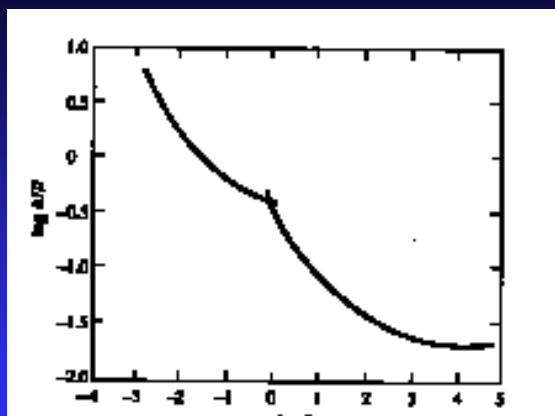


Figure 2.6 Typical Weber ratio as a function of intensity.
(Adapted from Graham [1965].)

Contouring Effect

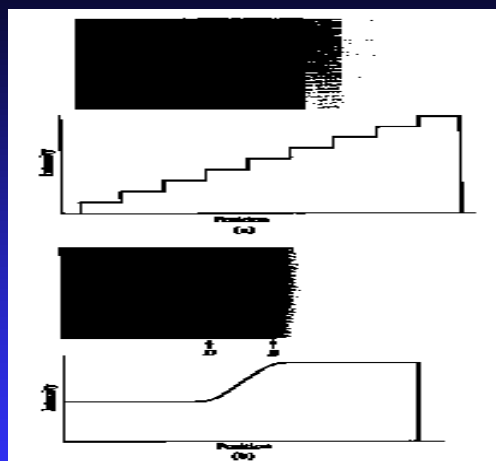


Figure 2.7 Two examples showing that perceived brightness is not a simple function of intensity.
(Adapted from Cornsweet [1970].)

Simultaneous Contrast

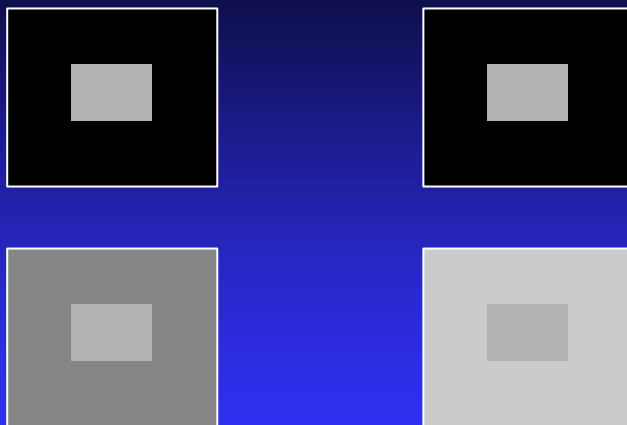


Figure 2.8 Example of simultaneous contrast

Simple Image Model

$f(x,y)$: two dimensional light-intensity function

$$0 < f(x,y) < \infty$$

$$f(x,y) = i(x,y)r(x,y)$$

$$0 < i(x,y) < \infty \quad \text{i: Illumination}$$

$$0 < r(x,y) < 1 \quad \text{r: Reflectance}$$

$$L_{\min} \leq \ell \leq L_{\max} \quad \text{l: gray scale}$$

$$[0,L] \quad \text{0: black, L: white}$$

Uniform Sampling and Quantization

$$f(x,y) = \begin{bmatrix} f(0,0) & f(0,1) & \dots & f(0,M-1) \\ f(1,0) & f(1,1) & \dots & f(1,M-1) \\ \vdots & \vdots & \ddots & \vdots \\ f(N-1,0) & f(N-1,1) & \dots & f(N-1,M-1) \end{bmatrix} N \times M$$

$$N = 2^n, M = 2^k, G = 2^m$$

$$b = N \times M \times m = N^2 m \quad (\text{if } N=M)$$

G: number of gray levels

b: bits required to store a digitized image

How many samples and gray levels are required for good approximation? (resolution)

Table of Storage Bits

m \ N	1	2	3	4	5	6	7	8
32	1,024	2,048	3,072	4,096	5,120	6,144	7,168	8,192
64	4,096	8,192	12,288	16,384	20,480	24,576	28,672	32,768
128	16,384	32,768	49,152	65,536	81,920	98,304	114,688	131,072
256	65,536	131,072	196,608	262,144	327,680	393,216	458,752	524,288
512	262,144	524,288	786,432	1,048,576	1,310,720	1,572,864	1,835,008	2,097,152
1,024	1,048,576	2,097,152	3,145,728	4,194,304	5,242,880	6,291,456	7,340,032	8,388,608

Table 2.1 Number of Storage Bits of Various Values of N and M

m \ N	1	2	3	4	5	6	7	8
32	128	256	512	512	1,024	1,024	1,024	1,024
64	512	1,024	2,048	2,048	4,096	4,096	4,096	4,096
128	2,048	4,096	8,192	8,192	16,384	16,384	16,384	16,384
256	8,192	16,384	32,768	32,768	65,536	65,536	65,536	65,536
512	32,768	65,536	131,072	131,072	262,144	262,144	262,144	262,144
1,024	131,072	262,144	393,216	524,288	655,360	786,432	917,504	1,048,576

Table 2.2 Number of 8-bit Bytes of Storage for Various Values of N and M

Effects of Variation of N and level

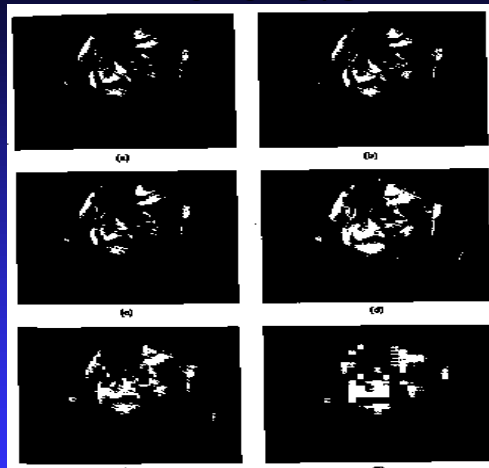


Figure 2.9 Effects of reducing spatial resolution.

Effects of Variation of N and level

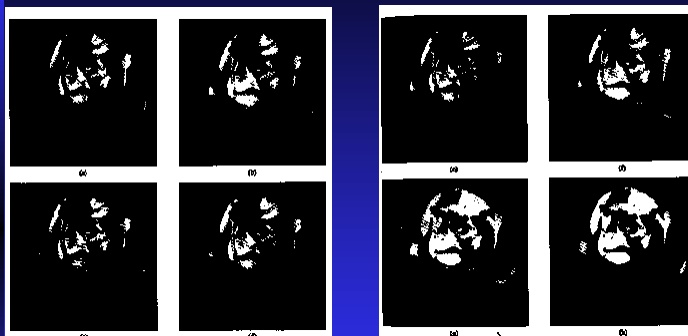


Figure 2.10 A 1024x1024 image displayed in 256, 128, 64, 32, 16, 8, 4, and 2 levels, respectively

Isopreference curves

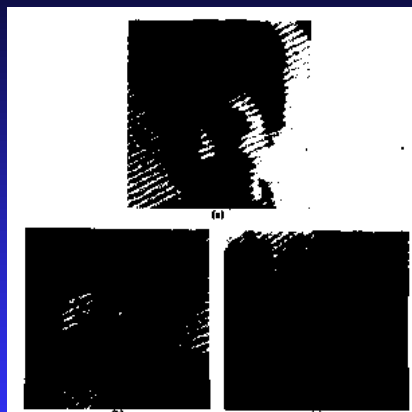


Figure 2.11 Test images used in evaluating subjective image quality. (From Huang [1965].)

Isopreference curves

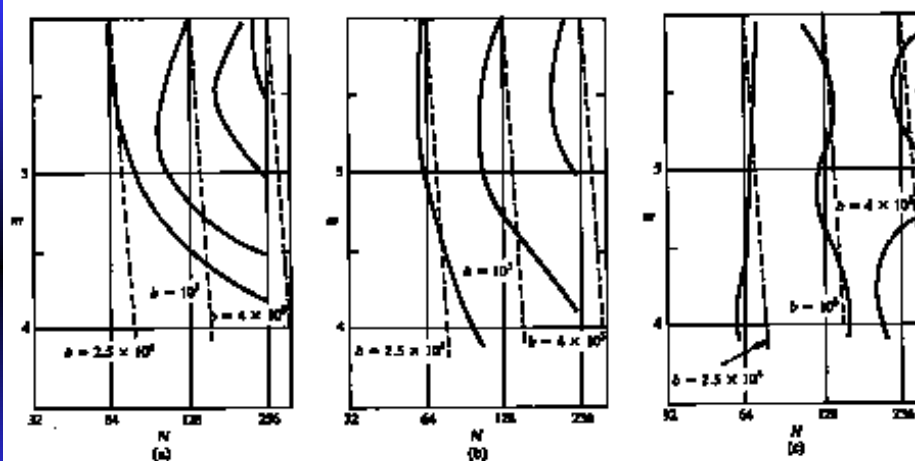
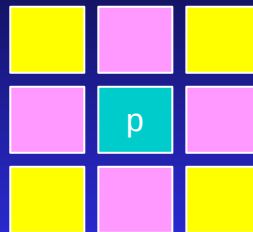


Figure 2.12 Isopreference curves for (a) face, (b) camera man, and (c) crowd. (From Huang [1965].)

Nonuniform Sampling and Quantization

1. Sharp gray level transition - fine sampling
Smooth gray level transition - coarse sampling
2. Number of gray levels **small** - unequally spaced level.
3. boundary - few gray level
smooth - remain level can be used
4. tapered quantization - computing the freq.
of occurrence of all allowed levels

Neighbors of a Pixel



: 4-neighbors of p $N_4(p)$



: diagonal neighbors of p $N_D(p)$

$$N_D(p) + N_4(p) = N_8(p)$$

Connectivity

4-connectivity : p and q from V , and q is in the set $N_4(p)$

8-connectivity : p and q from V , and q is in the set $N_8(p)$

m -connectivity : p and q from V , and q is m -connected if

i) q is in $N_4(p)$ or

ii) q is in $N_D(p)$

and the set $N_4(p) \cap N_4(q)$ is empty.

Path

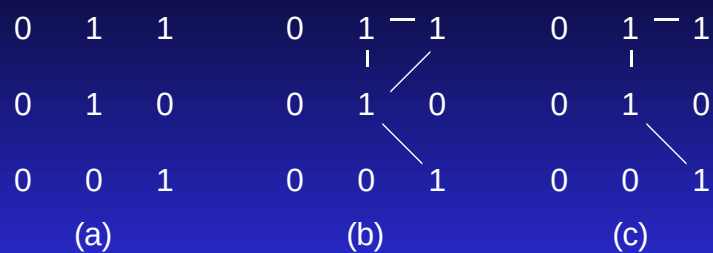


Figure 2.13

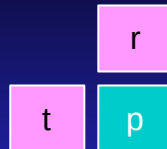
(a) Arrangement of pixels

(b) 8-neighbors of the center pixel

(c) m -neighbors of the same pixel

Labeling of Connected Components

4-connected:

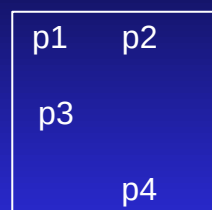


8-connected:



Relations

R : binary relation $\rightarrow$ (a,b) pair : aR_b



$A = \{ p1, p2, p3, p4 \}$

4-connected : $R = \{(p1, p2), (p2, p1), (p1, p3), (p3, p1)\}$

(a) reflexive : aR_a

(b) symmetric : $aR_b \rightarrow bR_a$

(c) transitive : aR_b and $bR_c \rightarrow aR_c$

Transitive Closure

$$R = \{(a,a), (a,b), (b,d), (d,b), (c,e)\}$$

$$B = \begin{matrix} & \begin{matrix} a & b & c & d & e \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \\ e \end{matrix} & \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

$$R^+ = \{(a,a), (a,b), (a,d), (b,b), (b,d), (d,b), (d,d), (c,e)\}$$

$$B^+ = \begin{matrix} & \begin{matrix} a & b & c & d & e \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \\ e \end{matrix} & \begin{bmatrix} 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

Generate Transitive Closure

$$1. \quad B^+ = B + BB + BBB + \dots + (B)^n$$

(Gries [1971].)

2. Warshall's Algorithm

step1. set $j = 1$

step2. For $i = 1, 2, \dots, n$, if $b(i, j) = 1$, then, for $k = 1, 2, \dots, n$,
set $b(i, k) = b(i, k) + b(j, k)$.

step3. Increment j by 1

step4. If $j \leq n$, goto step2; otherwise goto step5

step5. Stop

Distance Measures

Properties $p(x,y), q(s,t), z(u,v)$

(a) $D(p,q) \geq 0$ ($D(p,q) = 0$ iff $p=q$)

(b) $D(p,q) = D(q,p)$

(c) $D(p,z) \leq D(p,q) + D(q,z)$

Euclidean distance: $D_e(p,q) = [(x-s)^2 + (y-t)^2]^{1/2}$

City-block distance: $D_4(p,q) = |x-s| + |y-t|$

Chessboard distance: $D_8(p,q) = \max(|x-s|, |y-t|)$

Arithmetic/Logic Operation

Addition: $p + q$

Subtraction: $p - q$

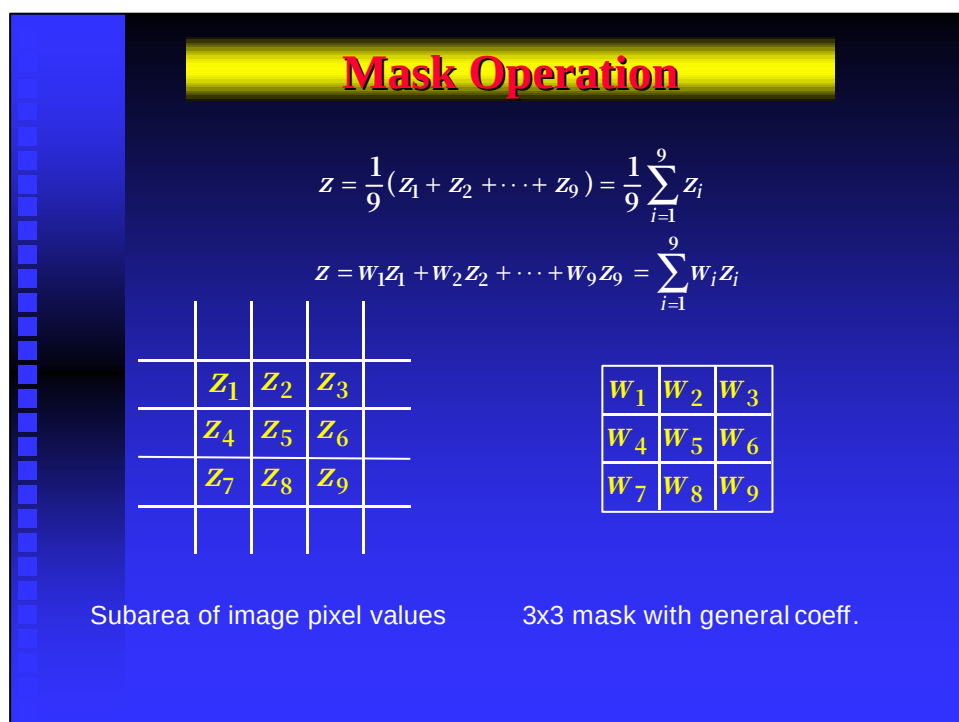
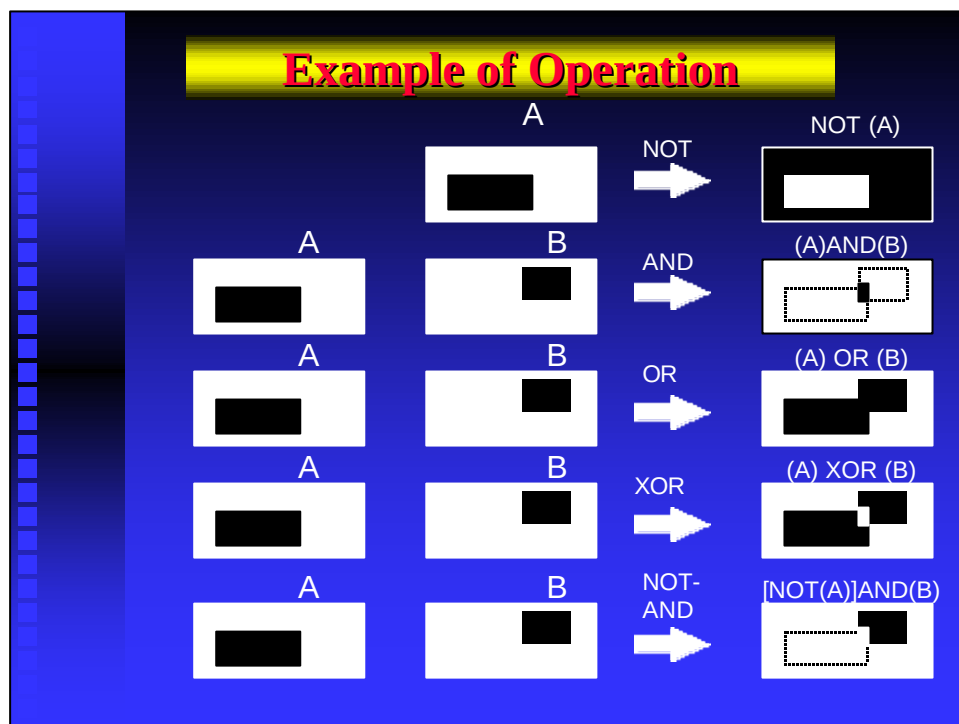
Multiplication: $p * q$ ($p \cdot q, p \times q$)

Division: $p \div q$

AND: $p \text{ AND } q$ ($p \cdot q$)

OR: $p \text{ OR } q$ ($p + q$)

COMPLEMENT: NOT q ($\bar{q}$)



ALU Operation	
Operations on A	Operations on B
-	Multiply by w5
Shift right	-
-	Add w4 * A
Shift down	-
-	Add w1 * A
Shift left	-
-	Add w2 * A
Shift left	-
-	Add w3 * A
Shift Up	-
-	Add w6 * A
Shift Up	-
-	Add w9 * A
Shift right	-
-	Add w8 * A
Shift right	-
-	Add w7 * A
Shift left	-
Shift down	-

Imaging Geometry	
Translation	$\mathbf{v}^* = \mathbf{A} \mathbf{v}$ $\begin{bmatrix} X^* \\ Y^* \\ Z^* \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & X_0 \\ 0 & 1 & 0 & Y_0 \\ 0 & 0 & 1 & Z_0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$
Scaling	$\mathbf{S} = \begin{bmatrix} S_x & 0 & 0 & 0 \\ 0 & S_y & 0 & 0 \\ 0 & 0 & S_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ ↙ T
Rotation	
$R_b = \begin{bmatrix} \cos b & 0 & -\sin b & 0 \\ 0 & 1 & 0 & 0 \\ \sin b & 0 & \cos b & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	$R_q = \begin{bmatrix} \cos q & \sin q & 0 & 0 \\ -\sin q & \cos q & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
	$R_a = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos a & \sin a & 0 \\ 0 & -\sin a & \cos a & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

Imaging Geometry

$$T^{-1} = \begin{bmatrix} 1 & 0 & 0 & -X_0 \\ 0 & 1 & 0 & -Y_0 \\ 0 & 0 & 1 & -Z_0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R_q^{-1} = \begin{bmatrix} \cos(-q) & \sin(-q) & 0 & 0 \\ -\sin(-q) & \cos(-q) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Perspective Transformation

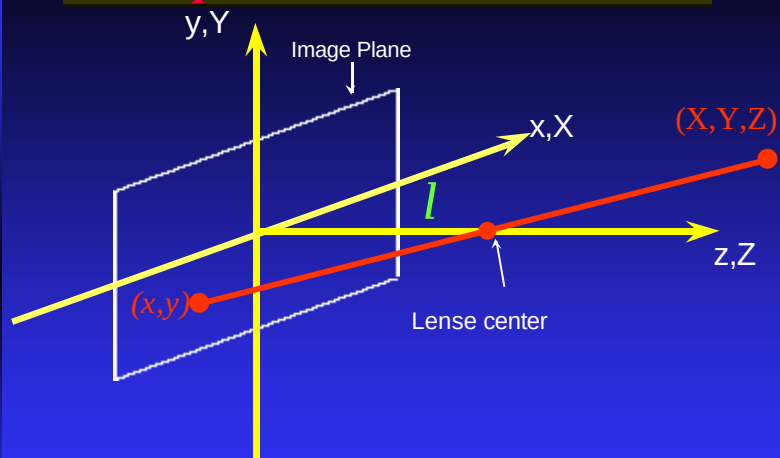


Figure 2.17 Basic Model of the imaging process

Perspective Transformation

$$\frac{x}{l} = -\frac{X}{Z-l} = \frac{X}{l-Z}$$

$$\frac{y}{l} = -\frac{Y}{Z-l} = \frac{Y}{l-Z}$$

$$x = \frac{lX}{l-Z}, y = \frac{lY}{l-Z}$$

$$c_h = Pw_h = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -\frac{1}{l} & 1 \end{bmatrix} \begin{bmatrix} kX \\ kY \\ kZ \\ k \end{bmatrix} = \begin{bmatrix} kX \\ kY \\ kZ \\ -\frac{kZ}{l} + k \end{bmatrix}$$

$$c = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \frac{lX}{l-Z} \\ \frac{lY}{l-Z} \\ \frac{lZ}{l-Z} \end{bmatrix}$$

Perspective Transformation

$$w_h = P^{-1}c_h = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & \frac{1}{l} & 1 \end{bmatrix} \begin{bmatrix} kx_0 \\ ky_0 \\ 0 \\ k \end{bmatrix}$$

$$w = \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} \frac{kx_0}{l+z} \\ \frac{ky_0}{l+z} \\ \frac{lz}{l+z} \end{bmatrix}$$

$$X = \frac{x_0}{l} (l-Z)$$

$$Y = \frac{y_0}{l} (l-Z)$$

Camera model

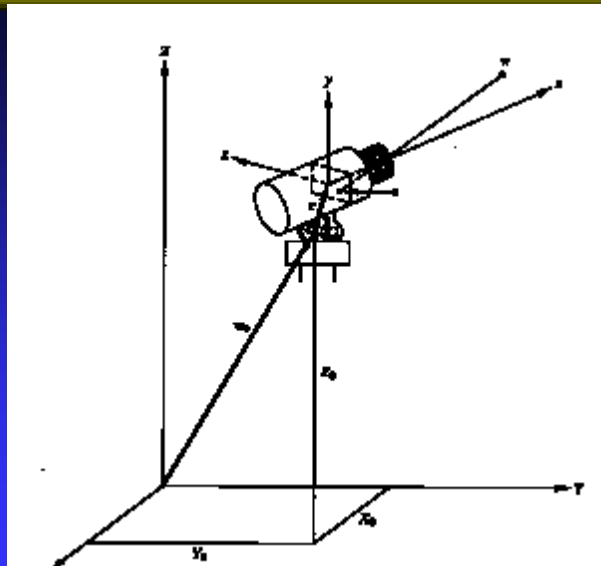


Figure 2.18 Imaging geometry with two coordinate systems

Camera model

pan : x 와 X 사이의 각

tilt: z 와 Z 사이의 각

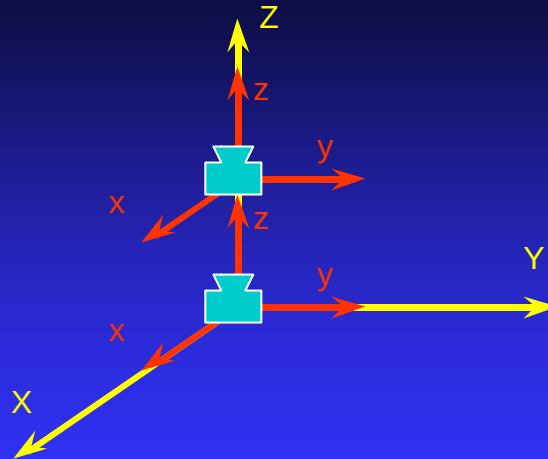
$$c_h = P C R G w_h$$

$$R = R_a R_q = \begin{bmatrix} \cos q & \sin q & 0 & 0 \\ -\sin q \cos a & \cos q \cos a & \sin a & 0 \\ \sin q \sin a & -\cos q \sin a & \cos a & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

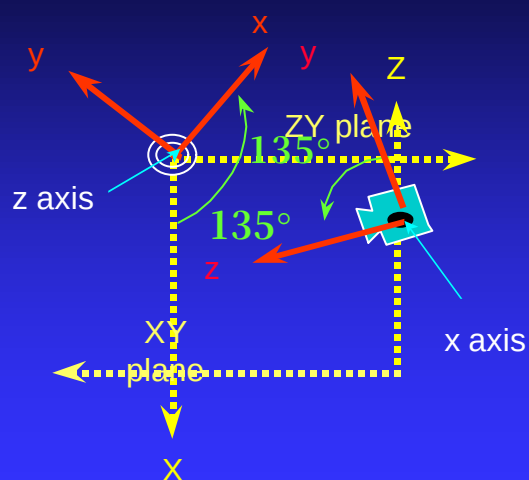
$$C = \begin{bmatrix} 1 & 0 & 0 & -r_1 \\ 0 & 1 & 0 & -r_2 \\ 0 & 0 & 1 & -r_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$G = \begin{bmatrix} 1 & 0 & 0 & -X_0 \\ 0 & 1 & 0 & -Y_0 \\ 0 & 0 & 1 & -Z_0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Example of Camera model (1)



Example of Camera model (2)



Camera Calibration

$$A = PCRG$$

$$c_h = Aw_h$$

$$\begin{bmatrix} c_{h1} \\ c_{h2} \\ c_{h3} \\ c_{h4} \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

$$a_{11}X + a_{12}Y + a_{13}Z - a_{41}X - a_{42}Y - a_{43}Z - a_{44}X + a_{14} = 0$$

$$a_{21}X + a_{22}Y + a_{23}Z - a_{41}Y - a_{42}Y - a_{43}Z - a_{44}Y + a_{24} = 0$$

Stereo Imaging

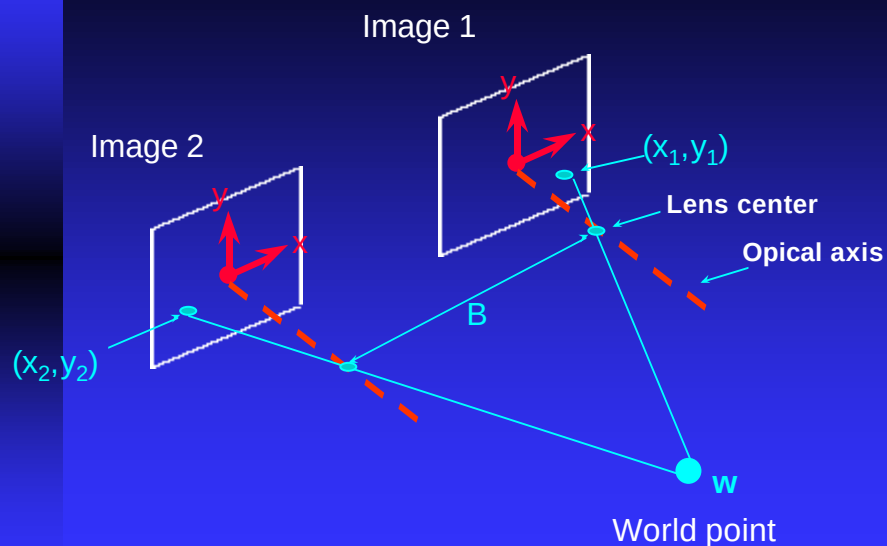
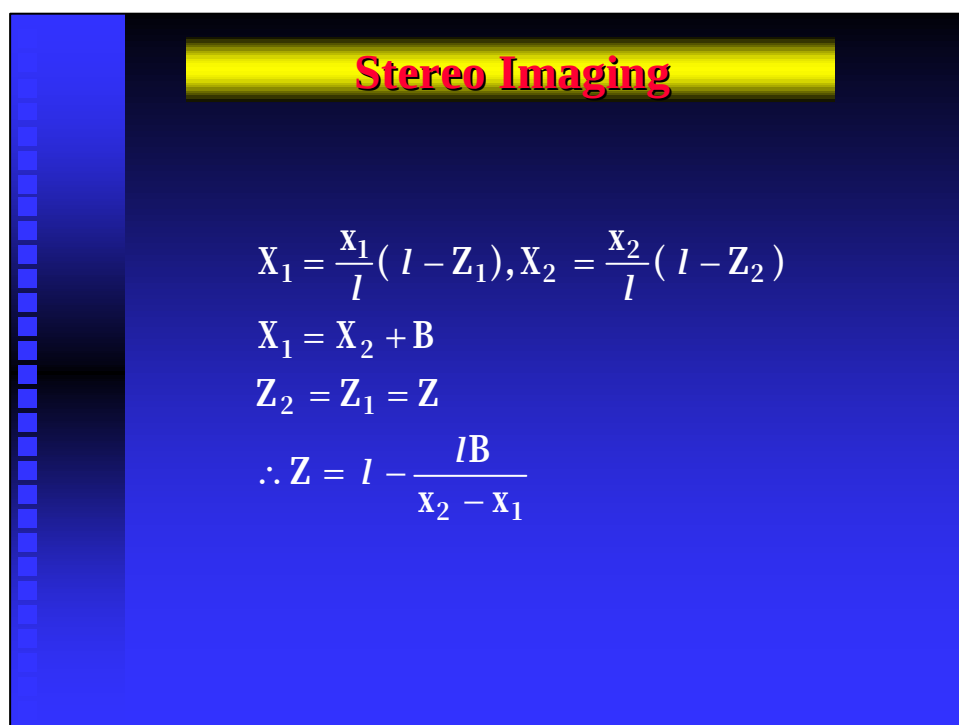
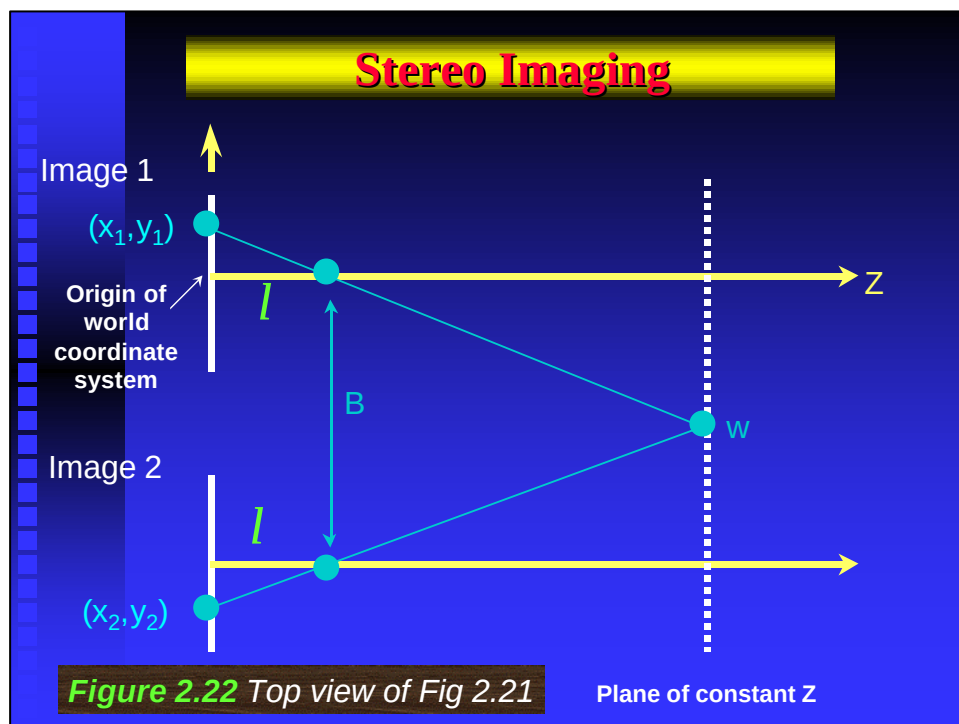


Figure 2.21 Model of the stereo imaging process



Photographic Film



Figure 2.23 Structure of modern black-and-white film

Photographic Film

Contrast

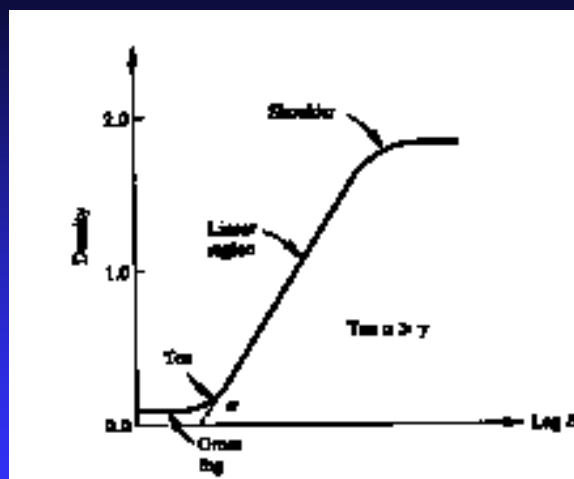


Figure 2.24 A typical H & D curve

Photographic Film

Speed

ASA scale : arithmetic
DIN scale : logarithmic

ASA50 = DIN18, ASA100 = DIN 21

Graininess

fast -> large crystal
slow -> fine crystal

Resolving Power

Fine-grain, thin emulsion -> highest resolving power

Photographic Film

lens diaphragm

f-number : 들어오는 빛의 양에 반비례
Shutter speed setting

In this book

1/4 sec Kodak Panatomic - X fine grain film (ASA32)